Acumatica MCP Localhost Integration Guide

Exposing Local IIS Instances to Cloudflare Worker MCP Server via Cloudflare Tunnel

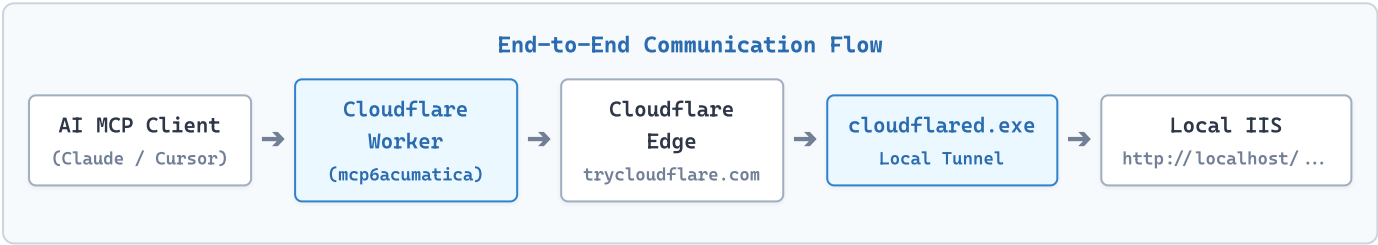
CLOUDFLARE TUNNEL | MCP SERVER | ACUMATICA XRP

Environment: Local Dev IIS / Cloudflare Worker
Protocol: HTTPS / WSS / OIDC OAuth 2.0
Version: Acumatica 2024R2+ / 2025R1+

Overview & Problem Statement

When testing or developing Acumatica customizations (such as PMS, custom DACs, or Generic Inquiries) on your local workstation, your instance runs inside local IIS (e.g., `http://localhost/Acumatica26R2/`).

The **Model Context Protocol (MCP) Server** runs as a Cloudflare Worker in the public cloud. Cloudflare Workers cannot directly call private loopback addresses (`localhost` or private LAN IPs). **Cloudflare Tunnel (cloudflared)** creates an outbound, encrypted reverse tunnel from your local machine to Cloudflare's global edge network, providing a public HTTPS URL without modifying firewall rules or forwarding router ports.



Prerequisites

- ✓ **Local Acumatica Instance:** Running on IIS and accessible in your local browser (e.g., `http://localhost/Acumatica26R2/`).
- ✓ **Node.js or cloudflared:** Node.js (v18+) installed to run `npx cloudflared`, or the standalone `cloudflared.exe` binary.
- ✓ **Deployed Cloudflare Worker:** Your MCP Worker deployed on Cloudflare (e.g., `mcp6acumatica.inglerohan1808.workers.dev`).
- ✓ **Administrator Access:** Access to Acumatica screen **Connected Applications (SM303010)**.

1 Launch Cloudflare Tunnel on Localhost

Open a PowerShell terminal on your Windows machine and execute the Quick Tunnel command:

```
# Start temporary Quick Tunnel pointing to local IIS port 80
npx cloudflared tunnel --url http://localhost
```

Cloudflare will establish a secure connection and output connection details:

```
+--------------------------------------------------------------------------------------------+
|  Your quick Tunnel has been created! Visit it at (it may take some time to be reachable):  |
|  https://announcement-auction-classic-magazines.trycloudflare.com                          |
+--------------------------------------------------------------------------------------------+
```

Verify Tunnel Connectivity & OIDC Endpoint

Open the tunnel URL combined with your Acumatica instance name and IdentityServer path in your browser or PowerShell: `https://<subdomain>.trycloudflare.com/Acumatica26R2/identity/.well-known/openid-configuration`
It must return a valid JSON configuration containing `authorization_endpoint` and `token_endpoint`.

Important: Keep the Terminal Open

Quick Tunnels remain active only while the PowerShell terminal window is running. If you close the terminal or press `Ctrl+C`, the tunnel terminates.

2 Configure Acumatica Connected Applications (SM303010)

Acumatica's built-in IdentityServer verifies redirect URIs during OAuth authorization code flow. Configure the client credentials as follows:

1. Log into your local Acumatica instance (`http://localhost/Acumatica26R2/`).
2. In the search box, open **Connected Applications** (Screen ID: `SM303010`).
3. Select or create an application for MCP (e.g., Client Name: `MCP Client`).
4. Set **OAuth Flow** to `Authorization Code`.
5. Click **Secrets → New Secret**, and securely copy the generated Secret.
6. Switch to the **REDIRECT URIS** tab and click **+**:

Setting	Required Value	Notes
Redirect URI	`https://mcp6acumatica.inglerohan1808.workers.dev/callback`	Must match your Cloudflare Worker's callback path exactly.
Scopes	`api`, `offline_access`	Enables REST API access and token refresh capabilities.
Client ID	`6D2DCD41-09F9-6EB8-6147-ED52F132EFE2@Company`	Includes the tenant identifier suffix (`@Company`).

3 Update Cloudflare Worker Variables & Secrets

Now inform your Cloudflare Worker where your Acumatica instance lives and provide its credentials:

1. Open the **Cloudflare Dashboard** (dash.cloudflare.com).
2. Navigate to **Workers & Pages →** Select your Worker (**mcp6acumatica**).
3. Go to **Settings → Variables and secrets**.
4. Configure the three critical environment variables:

Variable Name	Type	Configured Value
`ACUMATICA_URL`	Plaintext Variable	`https://<subdomain>.trycloudflare.com/Acumatica26R2` *Example:* `https://announcement-auction-classic-magazines.trycloudflare.com/Acumatica26R2`
`ACUMATICA_CLIENT_ID`	Plaintext Variable	`6D2DCD41-09F9-6EB8-6147-ED52F132EFE2@Company`
`ACUMATICA_CLIENT_SECRET`	Encrypted Secret	*(Your copied Secret string from SM303010)*

> **Save & Deploy**
> Click **Deploy** (or **Save and Deploy**). The Cloudflare Worker instantly updates its runtime environment across all global points of presence without downtime.

4 Register MCP in AI Clients & Authorize

Add your MCP Server to your AI client configuration (Claude Desktop, Cursor, or Antigravity):

```
// claude_desktop_config.json or cursor settings:
{
  "mcpServers": {
    "acumatica-local": {
      "url": "https://mcp6acumatica.inglerohan1808.workers.dev/sse"
    }
  }
}
```

First-Time OAuth Authorization:

1. When the AI client initiates connection, an Acumatica OAuth login prompt will launch.
2. Notice the browser opens the secure tunnel URL: `https://<subdomain>.trycloudflare.com/ ...`
3. Enter your local Acumatica credentials (e.g. `admin`) and click **Authorize**.
4. Acumatica redirects to the Cloudflare Worker callback, securely exchanging the authorization code for an OAuth token stored encrypted in Cloudflare KV.

5 Troubleshooting & Best Practices

Issue	Root Cause	Remediation
Tunnel URL changes after restart	Quick Tunnels (`npx cloudflared`) generate random ephemeral domains on every run.	Update `ACUMATICA_URL` in Cloudflare Worker settings upon restart, or create a permanent Named Tunnel using a free Cloudflare domain.
HTTP 502 Bad Gateway from Tunnel	Local IIS service stopped or binding is not listening on `http://localhost:80`.	Run `iisreset` or verify in IIS Manager that the Default Web Site binding is active on port 80.
OAuth Redirect URI Mismatch	The callback URL in SM303010 does not match the Worker domain.	Verify exact spelling of `https://<worker>.workers.dev/callback` with no trailing slashes or spaces.
401 Unauthorized during GI Queries	Acumatica user lacks permissions for generic inquiry or Canary GI.	Ensure the user account has access to the `MCPAccess` canary inquiry and target screens.